Game Name: The Last Oasis

Final Game Project Report

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1. Game Overview

The Last Oasis is a pixel-art, rogue-like action game built in Unity. Set in a futuristic, dystopian city, players choose between two characters and engage in fast-paced dungeon exploration. The goal is to survive waves of enemies, collect gold, and upgrade stats like health and attack power, all while unraveling the world through interactions and narrative encounters.

2. Core Gameplay Loop

- **Movement**: Players control characters using the **WASD** keys.
- **Combat**:
  - Light attack: **Left Mouse**
  - Heavy attack: **Right Mouse**
- **Interaction**:
  - **E** to interact (Exit label Use E to next level)
  - **Space** to progress through dialogues

Players collect **gold** through defeating enemies or exploring chests. Gold is then used to **upgrade health and attack** stats, creating a compelling **risk-reward** loop where survival leads to greater strength.

3. Art, Aesthetics, & Sound Design

- **Visual Style**: Pixel-art aesthetic with retro-futuristic backgrounds and cyberpunk UI elements. The overall tone is dark and chaotic with dim lighting that immerse players in a world on the edge of collapse. The background environments are deliberately cluttered such as broken machines to show apocalyptic scenes.
- **UI Design**: Clear navigation buttons and explanatory text in the lower right corner of the game, and both menu and instruction help players quickly understand the game's settings, with consistent visual cues across screens (e.g., glowing neon buttons, clean typography). The room interface should be as rich as possible, with objects that reflect the background. And the player's gold and blood defenses in the lower left corner help players get better feedback on themselves in real time
- **Sound Design**: Sound Design: Includes volume control; background music to enhance the tension of the battle. Battles and interactions are accompanied by sound effects to enhance immersion. For example, there is a prompt sound for collecting gold coins, and there is also a sound for a successful hit to help players determine whether the attack is successful or not, not simply from the blood level bar.

Despite using a relatively minimalist pixel style, the city background and character sprites evoke a strong cyber-themed atmosphere

4. Narrative & Systems Integration

The game introduces **dialogue elements** and interactions with NPCs to deepen the narrative in the first room of the game. Though story details are minimal in the current prototype, there are hints of world-building through:

- City backdrop implying a post-apocalyptic or controlled environment-abandoned equipements and mutated organisms.
- NPC dialog is brief and purposeful in order to provide background information, convey the tone of the world, and facilitate scene transitions. For example, an NPC might say a few cryptic lines about “the last oasis” to give the player the goal of the game.
  Encounter dialogues that can evolve into quests or lore delivery
- Character-driven storytelling potential (two protagonists).The small difference that the two characters differ in their weapon choices nevertheless enhances the feel of the game and the player's expressiveness. It also helps to enrich the playability of the game.

The integration of **upgrades through combat and gold collection** ties gameplay progression into the world setting—survival and resource gathering as necessities in a ruined world.

5. Progression & Balance

- Gold-Based Progression: Killing enemies and finding chests increases the player’s gold, which can be spent on upgrades.
- Stat Balancing: Both health and attack power can be upgraded, giving players meaningful decisions based on their play style (tankier vs. aggressive builds).
  Item Interactions: Certain items like hamburgers restore HP, rewarding exploration and providing mid-combat recovery.
- Random Reward: Some gold drops grant small stat boosts (e.g., +1 HP, +1 ATK), offering micro-progression and encouraging exploration while subtly enhancing run-to-run balance.

The balancing is kept intentionally simple to support quick game loops typical of rogue-likes while giving enough feedback to show progression.

6. Playtesting & Iteration

- Based on playtesting, controls were tuned for responsiveness and clarity (e.g., updated movement and attack speeds).
- UI layouts were refined to be intuitive, such as the centered start interface and color-coded buttons.
- Difficulty was adjusted by scaling enemy health and gold drops to maintain consistent challenge across sessions.
- Minor bugs with character collision and health regeneration were fixed during testing cycles.

Playtesting & Iteration Record:

Test Time	Test Method	Participant	Feeback	Modify
2.10	self-testing	Team members	Poor map structure and unclear Room design	Optimize map layout to enhance visual differences and structural coherence between rooms
2.24	self-testing	Team members	Confusing room generation and unstable logic	Realization of rules for linking room structures and wall openings to form a controlled and varied room generation system
3.15	Friend Trial Feedback	Roomates	Fewer interactive elements; poor	Improve camera following logic;

			camera following;	Add collision interaction(BoxCollider2D)
3.25	Friend Trial Feedback	Roomates	Operational experience is not smooth; lack of clear prompts	Add operation guide hints (UI text/button blinking); adjust attack and movement feedback to improve feel
4.10	TA Mark Feedback	TA	Monsters standing still and not moving affects the pace of the game	Implementation of basic monster AI patrol to enhance the dynamic experience
4.11	self-testing	Team members	Lack of score records or leaderboards reduces the fun of repeated play and comparison	Add score statistics system and save simple leaderboards.
4.15	Friend Trail feedback	External player who are not of this course (2)	Single character limits gameplay fun	Add a second character with a different weapon style to increase strategy differentiation

				and replay value
4.17	Self testing Debug	Team members	Bug when restarting after exiting the game, the game state is not reset and you cannot start a new game properly	Clears all player state, global variables on exit to ensure everything starts from the initial state when reloading the scene
4.18	Test	Classmates	Players felt unsure about effectiveness of upgrades due to lack of immediate feedback	Added immediate visual/audio cues when purchasing upgrades (e.g., upgrade animations or distinct audio feedback).
4.18	Self-testing	Team Members	The boss character did not chase the player, causing combat to feel passive and less engaging	Added tracking behavior to the boss, allowing it to actively pursue the player, significantly increasing combat tension.

4.19	Friend Test	Roommates (2)	Players were unable to see the boss's health during fights, causing confusion about their progress.	Added a visible, dedicated boss health bar at the top of the screen, clearly showing the boss's remaining HP.

7. Technical Implementation

- Developed in **Unity** using:
  - **Canvas UI**: Health bars, gold counters, upgrade menus, and dialog boxes are all built using the Unity UI canvas and displayed into screen space for responsive scaling.
  - **Tilemap System**: Used for efficient creation of 2D grid-based levels. Custom tiles efine walls and grounds.
  - **Cinemachine Camera**: Provides smooth, dynamic camera tracking and damping effects for better player tracking and improved visualization.
  - **Animator and Animation State:** Controls player and enemy state transition animations (idle, moving, attacking, dead).
  - **AudioMixer** for dynamic sound control
  - **Collider System**: Use BoxCollider2D, CircleCollider2D, and Rigidbody2D for precise movement, collision detection, and interactive environment response. For example, not going through walls and attacking touch and npc dialog.
  - **ScriptableObjects** for upgrade management

- Scene transitions were handled with Unity's **scene manager**, allowing smooth navigation between menu, instruction, score, and game scenes.
- Custom **scripts** control enemy AI(Enemy.cs), player stats, item pickups,room generation such as RoomGenerator.cs and interaction prompts.
- **Design Patterns:** Gameplay features like attack, health, and movement are modular scripts attached to different GameObjects for reusability and scalability. Applied in **GameManager** and **AudioManager** to ensure persistent control over global game states and background music.

8. High Score & Replayability

- The game features a **high score leaderboard**, encouraging replayability and competition. Players aim to surpass previous runs, a classic rogue-like mechanic that extends engagement.
- As a rogue-like game, each run resets character stats but introduces **random rewards** and **room layouts**, offering a fresh challenge every time.
- Gold drops with stat bonuses (+1 HP, +1 ATK) add **micro-progression**, enhancing moment-to-moment engagement

9. Conclusion

The Last Oasis combines simple mechanics with stylistic pixel visuals and rogue-like replayability. The game delivers fast action, meaningful upgrades, and a hint of narrative progression, making it suitable for both short sessions and high-score chasers.

With more iteration, deeper story integration and enemy variety can elevate the experience even further. The project showcases solid Unity fundamentals, UI/UX design, and iterative game development aligned with MDA and player engagement principles.